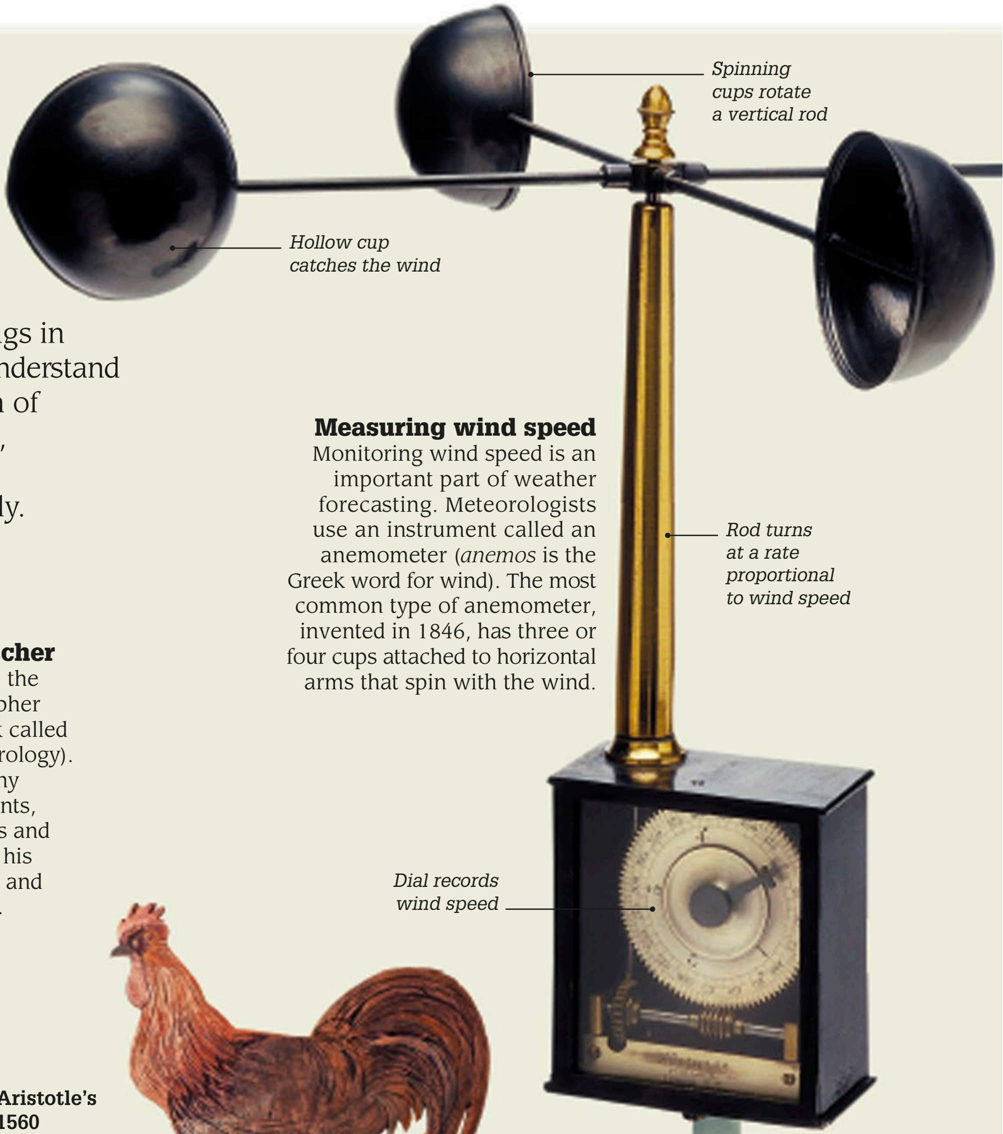
Studying weather

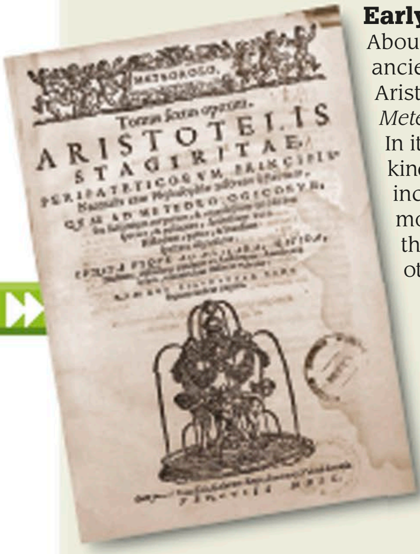
The science of weather and climate is called meteorology, a Greek word that originally meant the study of things in the sky. People have always tried to understand and predict the weather. The invention of instruments that measure air pressure, temperature, and humidity allow us to forecast weather more accurately.



Spinning-cup anemometer, c 1846

Measuring wind speed

Monitoring wind speed is an important part of weather forecasting. Meteorologists use an instrument called an anemometer (*anemos* is the Greek word for wind). The most common type of anemometer, invented in 1846, has three or four cups attached to horizontal arms that spin with the wind.



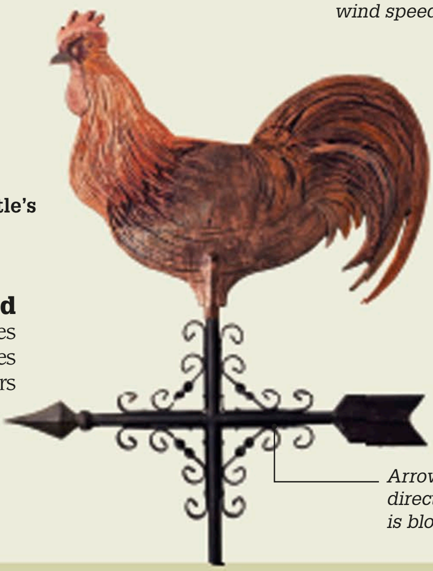
Latin version of Aristotle's *Meteorologica*, 1560

Early weather watcher

About 2,400 years ago, the ancient Greek philosopher Aristotle wrote a book called *Meteorologica* (Meteorology). In it he discussed many kinds of weather events, including whirlwinds and monsoons. Some of his theories were right, and others were wrong.

Turning in the wind

Fixed on the top of church steeples or other high buildings, weather vanes have been used for hundreds of years to show the direction of the wind—vital information for farmers, fishermen, and sailors.



Key events

1450

Leon Battista Alberti, an Italian architect, gave the first written description of a mechanical anemometer, an instrument used to measure the speed of wind.

1643

Italian physicist Evangelista Torricelli's experiments with vacuums and air pressure led to the development of the barometer, which measures changes in atmospheric pressure.

1686

English scientist Edmond Halley published a map charting the directions of ocean winds and monsoons. It is generally regarded as the first meteorological map.

1806

Francis Beaufort, an officer in the British Royal Navy, devised a scale for measuring wind speeds, which is still used today.